

Solution Methods for Second-Order ODEs

Abhishek Sarkar

IIT Jodhpur

Idea of the Method

- Consider the non-homogeneous ODE:

$$ay'' + by' + cy = g(x) \Rightarrow y'' + b_1 y' + c_1 y = g(x)$$

- General solution:

$$y = y_h + y_p$$

- y_h : solution of homogeneous equation

$$ay'' + by' + cy = 0$$

- y_p : particular solution (to be guessed)
- To get the final solution just add y_h and y_p .

Idea

- If g is a relatively simple function and the coefficients a , b and c are constants, then, after recalling what the derivatives of various basic functions look like, we might be able to make a good guess as to what sort of function $y_p(x)$ yields $g(x)$ after being plugged into the left side of the above equation.
- Typically, we won't be able to guess exactly what $y_p(x)$ should be, but we can often guess a formula for $y_p(x)$ involving specific functions and some constants that can then be determined by plugging the guessed formula for $y_p(x)$ into the differential equation and solving the resulting algebraic equation(s) for those constants (provided the initial 'guess' was good).

Example

EXR: Solve the same by V.O.F.

Example

Consider the non-homogeneous ODE and solve it:

$$y'' - 2y' - 3y = 36e^{5x}. \quad \text{--- (N)}$$

Reduced eqnⁿ: $y'' - 2y' - 3y = 0$. --- (H)

$$\Rightarrow y = c_1 e^{+3x} + c_2 e^{-x}$$

$$\Rightarrow r^2 - 2r - 3 = 0$$

$$\Rightarrow r = 3, -1$$

Guess:- $y_p = A e^{5x}$. --- Qn. $A = ?$

Differentiating $y_p' = 5A e^{5x}$ and $y_p'' = 25A e^{5x}$

value of y_p, y_p', y_p'' in (N) $\Rightarrow 25A e^{5x} - 10A e^{5x} - 3A e^{5x} = 36A e^{5x}$

$$y = c_1 e^{3x} + c_2 e^{-x} + 3e^{5x} \Rightarrow A = 3$$

Trial Forms for y_p

$g(x)$	Trial y_p: The Particular Solution
e^{ax}	Ae^{ax}
$\sin bx, \cos bx$	$A \cos bx + B \sin bx$
Polynomial x^n	$A_0 + A_1x + \cdots + A_nx^n$
$e^{ax}P_n(x)$	$e^{ax}(A_0 + \cdots + A_nx^n)$
$e^{ax} \sin bx, e^{ax} \cos bx$	$e^{ax}(A \cos bx + B \sin bx)$

Important Modification Rule

If the trial solution overlaps with y_h :

- Multiply the trial by x ✓
- If still overlapping, multiply by x^2 , etc.

Cases to be careful

Example

$$y'' - 2y' + y = e^x \quad \text{--- (N)}$$

- Complementary solution: $y_h = (C_1 + C_2x)e^x$
- Trial $y_p = Ae^x$ fails
- Use $y_p = Ax^2e^x$

$$y'' - 2y' + y = 0 \Rightarrow y_h(x) = (C_1 + C_2x)e^x.$$

Guess:- $y_p = Ae^x$

y_p will solve (N) $\Rightarrow Ae^x - 2Ae^x + Ae^x = e^x$

Guess: $y_p = Ax^2e^x \longrightarrow \Rightarrow \Leftarrow$

$$y_p = Ax^2 e^x$$

$$\Rightarrow y_p' = Ax^2 e^x + 2Ax e^x$$

$$y_p'' = Ax^2 e^x + 2Ax e^x + 2Ae^x + 2Ax e^x$$

$$= Ax^2 e^x + 4Ax e^x + 2Ae^x$$

$$y'' - 2y' + y = e^x$$

$$y_p'' - 2y_p' - y_p = e^x$$

$$\Rightarrow Ax^2 e^x + 4Ax e^x + 2Ae^x - 2Ax^2 e^x - 4Ax e^x - Ax^2 e^x = e^x$$

$$\Rightarrow \boxed{A = ?} \quad \checkmark$$

$$\text{--- } \boxed{\text{V.O.P}} \leftarrow \text{EXR}$$

Steps of the Method

- 1 Solve the homogeneous equation to get y_h
- 2 Guess a trial form for y_p
- 3 Modify if it overlaps with y_h
- 4 Substitute into the equation
- 5 Solve for coefficients

} Method of
undetermined
coefficients!

Method Selection for Non-Homogeneous ODEs

RHS $g(x)$	Method
Polynomial, e^{ax} , $\sin x$, $\cos x$	Undetermined Coefficients ✓
$\log x$, $\tan x$, $\frac{1}{x}$, etc.	Variation of Parameters ✓

Higher Order Linear ODEs

- We now turn our attention to solving linear differential equations of order n .

- The general form of such an equation is

$$a_0(x)y^{(n)} + a_1(x)y^{(n-1)} + \dots + a_{n-1}(x)y' + a_n(x)y = F(x)$$

variable

$$F(x, y, y', y'', \dots, y^{(n)}) = 0$$

general form

where $a_0, a_1, \dots, a_n$, and F are functions defined on an interval I .

- The general strategy is to reformulate the above equation as $Ly = F$, where L is an appropriate linear transformation. In fact, L will be a linear differential operator.

$$a_0 y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = F(x) \quad (N)$$

constants

$$[a_0, a_1, \dots, a_n \in \mathbb{R}]$$

$$a_0 y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = 0. \quad (H)$$

$$Ly = a_0 y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y.$$

y_1, y_2 solve $Ly = 0$.

$\Downarrow$

$c_1 y_1 + c_2 y_2$ will also solve $Ly = 0$

Since L is linear.

$$\dim(\text{Null}(L)) = n$$

Guess:- Solⁿ in the form of $\boxed{y = e^{rx}}$

$$y' = r e^{rx}$$

$$y^{(n)} = r^n e^{rx}$$

$$a_0 r^n e^{rx} + a_1 r^{n-1} e^{rx} + \dots + a_{n-1} r e^{rx} + a_n e^{rx} = 0$$

$$\Rightarrow e^{rx} (a_0 r^n + a_1 r^{n-1} + \dots + a_{n-1} r + a_n) = 0$$

Since $\boxed{e^{rx} \neq 0} \Rightarrow \boxed{a_0 r^n + a_1 r^{n-1} + \dots + a_{n-1} r + a_n = 0}$

$\hookrightarrow r$ will have n roots

Solution to the Linear Homogeneous ODE

- Consider the homogeneous linear differential equation

WLOG $a_0 = 1$

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} y' + a_n y = 0 = P(D)y$$

with constant coefficients a_i .

- Expressed as a linear differential operator, the equation is $P(D)y = 0$, where

$$P(D) = D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n$$

$$D \equiv \frac{d}{dx}$$

Definition

A linear differential operator with constant coefficients, such as $P(D)$, is called a polynomial differential operator. The polynomial

$$P(r) = r^n + a_1 r^{n-1} + \cdots + a_{n-1} r + a_n$$

is called the auxiliary polynomial, and the equation $P(r) = 0$ the auxiliary equation.

Solutions Based on The Roots

Theorem

The differential equation

$$(D - r)^m y = 0$$

has the following m linearly independent solutions:

$$e^{rx}, xe^{rx}, x^2 e^{rx}, \dots, x^{m-1} e^{rx}.$$

Handwritten notes and diagrams:

- n (with an arrow pointing to $P(D)y=0$)
- $m < n$ (boxed)
- $P(D)y=0$
- $m=3, 4, \dots, W.C. \neq 0$
- $e^{r_{m+1}x}, x e^{r_{m+1}x}$ (underlined)
- $r_0, \dots, r_m$ (grouped by a bracket)
- $r_{m+1}, \dots, r_n$ (grouped by a bracket)
- $(n-m)$ distinct
- $e^{r_0x}, x e^{r_0x}, \dots, x^{m-1} e^{r_0x}, e^{r_{m+1}x}, x e^{r_{m+1}x}, \dots, e^{r_nx}$

Solutions Based on The Roots

$n > m$

$n = \text{odd}$

Theorem

If $P(D)y = 0$ is a linear differential equation with real constant coefficients and $(D - r)^m$ is a factor of $P(D)$ with $r = a + ib$ and $b \neq 0$, then

- $P(D)$ must also have the factor $(D - \bar{r})^m$
- this factor contributes the complex solutions

$$e^{(a \pm bi)x}, xe^{(a \pm bi)x}, \dots, x^{m-1}e^{(a \pm bi)x}$$

- the real and imaginary parts of the complex solutions are linearly independent real solutions $x^k e^{ax} \cos bx$ and $x^k e^{ax} \sin bx$ for $k = 0, 1, \dots, m-1$.

$$\begin{array}{l} z' + az + by = 0 \\ y' = z \end{array} \quad \left| \quad \begin{array}{l} z' = az + by \\ y' = z \end{array} \right| \quad \begin{array}{l} y'' + ay' + by = 0 \\ y' = z \quad | \quad y'' = z' \end{array}$$

$$\begin{pmatrix} z' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} z \\ y \end{pmatrix}$$

Thank You

Questions?